```
!pip install nltk
import nltk
nltk.download('punkt_tab')
nltk.download('reuters')
```

```
Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (3.9.1)
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from nltk) (8.3.1)
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from nltk) (1.5.3)
Requirement already satisfied: regex>=2021.8.3 in /usr/local/lib/python3.12/dist-packages (from nltk)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from nltk) (4.67.3)
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.
[nltk_data] Downloading package reuters to /root/nltk_data...
[nltk_data]   Package reuters is already up-to-date!
True
```

```
from nltk.corpus import reuters
from nltk.tokenize import word_tokenize

documents = reuters.fileids()

text = reuters.raw(documents[0])
tokens = word_tokenize(text.lower())

print(tokens[:50])
```

```
['asian', 'exporters', 'fear', 'damage', 'from', 'u.s.-japan', 'rift', 'mounting', 'trade', 'friction
```


```
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```


```
suffix_rules = {
    "ing": "PROGRESSIVE",
    "ed": "PAST",
    "s": "PLURAL",
    "es": "PLURAL",
    "ly": "ADVERB"
}
```

```
class MorphFSA:

    def __init__(self, rules):
        self.rules = rules

    def analyze(self, word):
        for suffix in self.rules:
            if word.endswith(suffix):
                root = word[:-len(suffix)]
                return {
                    "word": word,
                    "root": root,
                    "morphology": self.rules[suffix]
                }

        return {
            "word": word,
            "root": word,
            "morphology": "ROOT"
        }
```

```python
fsa = MorphFSA(suffix_rules)

for word in tokens[:20]:
    result = fsa.analyze(word)
    print(result)
```

```
{'word': 'asian', 'root': 'asian', 'morphology': 'ROOT'}
{'word': 'exporters', 'root': 'exporter', 'morphology': 'PLURAL'}
{'word': 'fear', 'root': 'fear', 'morphology': 'ROOT'}
{'word': 'damage', 'root': 'damage', 'morphology': 'ROOT'}
{'word': 'from', 'root': 'from', 'morphology': 'ROOT'}
{'word': 'u.s.-japan', 'root': 'u.s.-japan', 'morphology': 'ROOT'}
{'word': 'rift', 'root': 'rift', 'morphology': 'ROOT'}
{'word': 'mounting', 'root': 'mount', 'morphology': 'PROGRESSIVE'}
{'word': 'trade', 'root': 'trade', 'morphology': 'ROOT'}
{'word': 'friction', 'root': 'friction', 'morphology': 'ROOT'}
{'word': 'between', 'root': 'between', 'morphology': 'ROOT'}
{'word': 'the', 'root': 'the', 'morphology': 'ROOT'}
{'word': 'u.s.', 'root': 'u.s.', 'morphology': 'ROOT'}
{'word': 'and', 'root': 'and', 'morphology': 'ROOT'}
{'word': 'japan', 'root': 'japan', 'morphology': 'ROOT'}
{'word': 'has', 'root': 'ha', 'morphology': 'PLURAL'}
{'word': 'raised', 'root': 'rais', 'morphology': 'PAST'}
{'word': 'fears', 'root': 'fear', 'morphology': 'PLURAL'}
{'word': 'among', 'root': 'among', 'morphology': 'ROOT'}
{'word': 'many', 'root': 'many', 'morphology': 'ROOT'}
```